

| Course code                                                                                                                                                                                                                                                                                                                                                                                                                                  | Course Title              | L                | T | P | C |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------|---|---|---|
| BCSE333L                                                                                                                                                                                                                                                                                                                                                                                                                                     | Statistical Inference     | 2                | 0 | 0 | 2 |
| Pre-requisite                                                                                                                                                                                                                                                                                                                                                                                                                                | NIL                       | Syllabus version |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                              |                           | 1.0              |   |   |   |
| Course Objectives                                                                                                                                                                                                                                                                                                                                                                                                                            |                           |                  |   |   |   |
| <div>1. To study statistical methods for hypotheses testing and solving inference problems.</div> <div>2. To interpret the results in a way that draws evidence-based and well-informed decisions from data.</div> <div>3. To derive conclusions from data and analyze its implications.</div>                                                                                                                                               |                           |                  |   |   |   |
| Course Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                              |                           |                  |   |   |   |
| At the end of the course, the student will be able to                                                                                                                                                                                                                                                                                                                                                                                        |                           |                  |   |   |   |
| <div>1. Understand the notion of a parametric model, point estimation of the parameters and properties of a good estimator.</div> <div>2. Learn the concept of interval estimation and confidence intervals.</div> <div>3. Understand and perform large-sample tests of hypotheses.</div> <div>4. Discuss nonparametric tests of hypotheses.</div> <div>5. Translate and correlate the statistical analysis into Statistical inference</div> |                           |                  |   |   |   |
| Module:1                                                                                                                                                                                                                                                                                                                                                                                                                                     | Introduction to Estimator | 4 hours          |   |   |   |
| Population, sample, parameter and statistic- Estimator, Estimate-characteristics of a good estimator – Unbiasedness- Consistency-Invariance property of Consistent estimator-Sufficient condition for consistency- Sufficiency- Factorization Theorem- Minimal sufficiency-Efficiency- Applications of Lehmann-Scheffe's theorem, Rao - Blackwell Theorem and applications. Bayesian Estimation.                                             |                           |                  |   |   |   |
| Module:2                                                                                                                                                                                                                                                                                                                                                                                                                                     | Point Estimation          | 5 hours          |   |   |   |
| Methods of point estimation- Maximum likelihood method (the asymptotic properties of ML estimators are not included), Large sample properties of ML estimator (without proof)- applications of MLE, Method of Minimum variance, method of moments, method of least squares, method of minimum chi-square.                                                                                                                                    |                           |                  |   |   |   |
| Module:3                                                                                                                                                                                                                                                                                                                                                                                                                                     | Interval Estimation       | 3 hours          |   |   |   |
| Confidence limits and confidence coefficient; Duality between acceptance region of a test and a confidence interval; Construction of confidence intervals for population proportion (small and large samples) and between two population proportions (large samples); Confidence intervals for mean and variance of a normal population; Difference between the mean and ratio of two normal populations.                                    |                           |                  |   |   |   |
| Module:4                                                                                                                                                                                                                                                                                                                                                                                                                                     | Testing of hypotheses     | 4 hours          |   |   |   |
| Types of errors, power of a test, most powerful tests; Neyman-Pearson Fundamental Lemma and its applications; Notion of Uniformly most powerful tests; Likelihood Ratio tests: Description and property of LR tests - Application to standard distributions.                                                                                                                                                                                 |                           |                  |   |   |   |
| Module:5                                                                                                                                                                                                                                                                                                                                                                                                                                     | Large sample tests        | 4 hours          |   |   |   |
| Large sample properties; Tests of significance (under normality assumption)- Test for a single population mean, proportion; Test for equality of two means, proportions; Test for variance, Test for correlation and Test for Regression.                                                                                                                                                                                                    |                           |                  |   |   |   |
| Module:6                                                                                                                                                                                                                                                                                                                                                                                                                                     | Small sample tests        | 4 hours          |   |   |   |
| Student's t-test, test for a population mean, equality of two population means, paired t-test, F-test for equality of two population variances; Chi-square test for goodness of fit, independence of attributes.                                                                                                                                                                                                                             |                           |                  |   |   |   |
| Module:7                                                                                                                                                                                                                                                                                                                                                                                                                                     | Non-parametric tests      | 4 hours          |   |   |   |
| Sign test, Wilcoxon Signed rank test, Median test, Wilcoxon-Mann-Whitney test, Run test and One sample Kolmogorov Smirnov test, Kruskal Wallis-H-test: Description, properties and applications.                                                                                                                                                                                                                                             |                           |                  |   |   |   |